Alan Tseng

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SUMMARY

Systems software engineer with cross-stack experience spanning Android platform integration, Linux kernel and firmware debugging, OpenBMC development, and Kubernetes-based infrastructure automation. M.Sc. from the Graduate Institute of Networking and Multimedia, National Taiwan University. Strong in C/C++, Python, and low-level system analysis using GDB, perf, and uftrace. Active open-source contributor and COSCUP 2025 speaker on OpenBMC debugging. Currently building AI agent and harness frameworks for Android platform engineering workflows.

SKILLS

Languages	C, C++, Python, Go, Shell, JavaScript / TypeScript
Systems & Kernel	Linux kernel debugging, Android AOSP, device tree, drivers (I2C, SPI, GPIO, ADC, UART), systemd, D-Bus, performance analysis (perf, ftrace, uftrace), GDB / gdbserver, core dump analysis
Firmware & Embedded	OpenBMC, Yocto, Buildroot, IPMI, Redfish, BMC platform bring-up, sensor stack porting
Infrastructure & Cloud	Kubernetes (kubeadm, Helm), Docker, PXE / DHCP / TFTP provisioning, CI/CD pipelines, container image optimization
AI & Automation	LLM agent frameworks, RAG retrieval pipelines, tool-use / function calling, prompt engineering, agent harness design
Web & Tooling	WebGPU, WebAssembly (WASM), Web APIs, browser-based benchmarking, REST API design, pytest

EXPERIENCE

Project System Integration Engineer – Android Platform Sep. 2025 – Present
MediaTek, Taipei, Taiwan

- Debug and root-cause Android platform issues across the kernel, HAL, and framework layers for the Amazon tablet product line; analyze AOSP build failures, runtime logs, and benchmark regressions to identify defects and drive fixes with software, firmware, and QA teams.
- Built a browser-based performance benchmarking suite using WebGPU, WebAssembly, and Web APIs to estimate SoC and GPU performance on locked commercial devices without root access, enabling cross-vendor comparison and competitive analysis on production tablet SKUs.
- Designed and executed thermal and low-power experiments on Android tablets; correlated benchmark scores with thermal throttling behavior and power-rail telemetry, producing technical reports consumed by product and platform teams.
- Prototyped an internal AI agent harness for Android BSP engineering – integrating RAG-style document retrieval, function-calling tool use, and reusable agent skills against internal technical documentation; received internal vAward for AI productivity and security contribution.
- Promoted secure adoption of AI-assisted development workflows (Claude Code-class assistants, MCP-based tool servers) across BSP engineering teams; authored guidelines on prompt safety and tool isolation.

Embedded Systems Engineer – BMC Firmware Dec. 2024 – Sep. 2025
Wiwynn, New Taipei City, Taiwan

- Developed and debugged OpenBMC firmware on Yocto and Buildroot-based hyperscale server platforms; owned platform bring-up tasks across multiple SKUs covering boot, sensors, services, and management interfaces.
- Ported and validated device tree, I2C, GPIO, ADC, systemd services, and D-Bus interfaces for new BMC platforms; integrated IPMI and Redfish endpoints and verified compliance against datacenter customer specifications.
- Built a cross-debugging workflow using gdbserver, core dump analysis, and live-attach techniques to triage BMC firmware crashes and service failures in production server environments; presented this workflow as a public talk at COSCUP 2025.
- Authored Python and pytest-based firmware validation tools, reducing manual sanity-check time and catching regressions before customer hand-off.

Server Software Engineer – Provisioning Automation Sep. 2024 – Dec. 2024
Pegatron, Taipei City, Taiwan

- Designed and implemented Kubernetes-based PXE provisioning workflows for rack-scale ARM and x86 server validation, automating Windows and Linux OS deployment using kubeadm, Helm, Docker, DHCP, and TFTP.
- Integrated low-level networking protocols (ARP, DHCP, TFTP) into the provisioning pipeline and instrumented stage-by-stage logging to make boot failures diagnosable without physical console access.
- Reduced container image and repository size from 663 MB to 2.4 MB (~99% reduction) by restructuring base images, eliminating redundant artifacts, and rewriting Dockerfiles with multi-stage builds.

Software Engineer Intern Jul. 2022 – Nov. 2022
LINE Corporation, Taipei City, Taiwan

- Reduced backend API development time by ~20% by introducing Swagger-driven code generation and standardizing API specification workflows across the B2B platform team.
- Supported Kubernetes deployments, DNS routing, and load balancing for cloud-native backend services; contributed to Nx monorepo tooling and data-driven development practices.
- Led a 5-engineer team to build an NFT merchant platform (Next.js + Spring Boot) during an internal hackathon, owning architecture decisions and integration milestones.

EDUCATION

M.Sc., Graduate Institute of Networking and Multimedia, College of Electrical Engineering & Computer Science, National Taiwan University Aug. 2021 – Feb. 2024 | GPA: 4.12/4.3
B.S., Computer and Communication Engineering, Ming Chuan University Sep. 2017 – Jun. 2021 | GPA: 3.99/4.0

SELECTED PROJECTS

- **CPU-Side Low-Bit LLM Inference Analysis (T-MAC / BitNet)** – Profiled inference performance using `perf` and `uftrace`; analyzed IPC, cache and TLB misses, page faults, and function-level execution behavior to identify optimization targets.
- **Linux Kernel Internals & Open-Source Contributions** – Contributed course materials on eBPF, profiling, and memory management; contributed to *Operating System in 1000 Lines*. wiki
- **Adaptive Driving Beam System** – Designed a nighttime vehicle-detection and adaptive-beam system using CycleGAN, YOLO, statistical analysis, and SVM; published in CVGIP 2020.
- **Technical Writing** – Maintain a TIL and blog covering embedded systems, Linux debugging, AI-assisted development, and platform engineering workflows. TIL | Blog

TALKS, HONORS AND LEADERSHIP

- **Talk** – *Debugging OpenBMC Firmware*, COSCUP 2025; covering OpenBMC cross-debugging, gdbserver, core dump analysis, and live-attach workflows on production BMC platforms.
- **Honors** – Research Excellence Award (NTU); Intel Andy Grove Scholarship; College Student Research Scholarship from Taiwan's Ministry of Science and Technology.
- **Leadership** – Lead, Google Developer Student Club; Leader, ACM Competitive Programming Club; volunteer staff at SITCON and COSCUP community events.